$\mathbb{C} \longrightarrow \mathbb{R}$

x^2+1 $\qquad$ x^2+1 — indecomposable

$=(x+i)(x-i)$

$6=2\times3$, $3=1\times3$ — prime (素数)

Def. $a\in\mathbb{Z}_{>0}$ is called to be a prime number if $a\mid bc$, then $a\mid b$ or $a\mid c$

RK1. a is prime iff a is

RK2. $\mathbb{Z}[\sqrt{-5}]=\{a+b\sqrt{-5}\mid a,b\in\mathbb{Z}\}$

$9=3\times3=(2+\sqrt{-5})(2-\sqrt{-5})$

if 3 is prime , $3\mid(2+\sqrt{-5})(2+\sqrt{-5})$ But $3\nmid 2+\sqrt{-5}$, $3\nmid 2-\sqrt{-5}$

so need fresh concept !

$\boxed{\text{推广. 素 prime} \longrightarrow \text{irreducible } (F[x],\ \mathbb{Z}[x])}$

FTOA (Fundemental theorem of Arithmetic 算术)

$\forall n\geq2$. $\exists p_i$ — Primes s.t. $n=\prod_{i=1}^{s}p_i$ (unique up to order)

Def. $F\subseteq\mathbb{C}$, $f(x)\in F[x]$, $\partial f\geq1$ Then $f(x)$ is irreducible if $f(x)=g(x)h(x)$,

then $\partial g=\partial f$ or $\partial h=\partial f$

RK. In the def above $g(x)=cf(x)$ or $h(x)=cf(x)$

FTOA (Algebra) $\Longleftrightarrow$ polynomials with degree 1 are the only irreducible over $\mathbb{C}$

Thm over $\mathbb{R}$, the irreducible polynomials are of the following forms :

$ax+b$ $(a\neq0)$; ax^2+bx+c with $b^2-ac<0$.

pf. $\forall f(x)\in\mathbb{R}[x]\subseteq\mathbb{C}[x]$

so $f(x)=a\prod_{i=1}^{s}(x-r_i)\prod_{j=1}^{t}(x-c_j)$ where $r_i\in\mathbb{R}$, $c_j\in\mathbb{C}\setminus\mathbb{R}$

Then $\exists\,\sigma\in S_t$ s.t. $\bar{c}_j=c_{\sigma(j)}$

$t=2q$ Then $f(x)=a\prod_{i=1}^{s}(x-r_i)\prod_{j=1}^{2q}\underbrace{(x-c_j)(x-c_{\sigma(j)})}_{x^2+\alpha_jx+\beta_j}$

Question: What are irreducible polynomials over $\mathbb{Q}$?

Example: $\forall n\geq1$ x^n+2 is irreducible over $\mathbb{Q}$

Eisenstein (爱森斯坦) criterion

Primitive Polynomial (本原多项式) over $\mathbb{Z}[x]$

$f(x) = a_0 + a_1 x + \cdots + a_n x^n \ (a_n \neq 0)$ with $(a_0, \cdots a_n) = 1$

Gaussian Lemma

Let $f(x), g(x) \in \mathbb{Z}[x]$ be two primitive polynomials, then $f(x)g(x)$ is also primitive

Eisenstein Criterion

Let $f(x) \in \mathbb{Z}[x]$, with $\partial f = n$, p is a prime number

$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$

s.t. (i) $p \mid a_i \quad 0 \le i \le n-1$

(ii) $p \nmid a_n$

(iii) $p^2 \nmid a_0$

Then $f(x)$ is irreducible over $\mathbb{Z}$.

Ex. Let p be prime, $1 + x + x^2 + \cdots + x^{p-1}$ is reducible?

$f(x) = \sum_{i=0}^{p-1} x^i = \frac{x^p - 1}{x-1}$

$\underline{\underline{x = y+1}} \quad \frac{(y+1)^p - 1}{y} = y^{p-1} + C_p^1 y^{p-2} + C_p^2 y^{p-3} + \cdots + p$

if p is prime $p \mid C_p^r \quad 1 \le r \le p-1$

so $p \mid C_p^1 y^{p-2} + C_p^2 y^{p-3} + \cdots + p$, but $p \nmid y^{p-1}$

so $1 + x + x^2 + \cdots + x^{p-1}$ is irreducible

Rk. $\mathbb{F}[x]$ is the best

c is a root of $f(x) \Leftrightarrow x - c \mid f(x)$

c is a multiple root of $f(x)$ with multiplicity $k \ge 2$ if $(x-c)^k \mid f(x)$

but $(x-c)^{k+1} \nmid f(x)$

Thm. Let $f(x) \in \mathbb{F}[x]$, Then $f(x)$ has a multiple root iff $(f(x), f'(x)) \neq 1$

$F[x] \xrightarrow{\partial} F[x]$

$$\begin{cases} \partial(af(x) + bg(x)) = a\,\partial f(x) + b\,\partial g(x) \\ \partial(f \cdot g) = \partial f \cdot g + f\,\partial g \\ \partial x = 1 \end{cases}$$